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XII-PQ-25 BIO

# Pre Quarterly Examination - 2025

## Standard XII

## BIOLOGY

Time : 3.00 hrs. Marks : 70

*Instructions: 1) Check the question paper for fairness of printing. If there is any lack of fairness, inform the Hall Supervisor immediately.*

*2) Use Black or Blue ink to write and underline and pencil to draw diagrams.*

*Note: Candidate should answer Part - I (Botany) & Part - II (Zoology) in separate answer - books.*

### PART - I (BIO - BOTANY)

### (Marks : 35)

### Section - A

*Note: i) Answer all the questions.* 8x1=8

*ii) Choose the most suitable answer from the given four alternatives and write the option code and the corresponding answer.*

1. The scar left by funiculus in the seed is
   a) tegmen b) radicle c) epicotyl d) hilum

2. The term genetics was introduced by .....
   a) Mendel b) Punnett c) Bateson d) Darwin

3. An allohexaploidy contains
   a) six different genomes
   b) six copies of three different genomes
   c) two copies of three different genomes
   d) six copies of one genomes

4. pBR 322, BR stands for
   a) Plasmid Bacterial Recombination
   b) Plasmid Bacterial Replication
   c) Plasmid Boliver and Rodriguerz
   d) Plasmid Baltimore and Rodriguerz

5. Identify the correctly matched pair.
   a) Tuber - *Allium cepa*
   b) Sucker - *Pistia*
   c) Rhizome - *Musa*
   d) Stolon - *Zingiber*

"O My Lord, Increase me in knowledge" - Holy Qur... 20:114

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6. Virus free plants are developed from
   a) Organ culture
   b) Meristem culture
   c) Protoplast culture
   d) Cell suspension culture

7. Ecology is the study of an individual species is called
   i) Community ecology
   ii) Autecology
   iii) Species ecology
   iv) Synecology
   a) (i) only
   b) (ii) only
   c) (i) and (iv) only
   d) (ii) and (iii) only

8. Pedogenesis refers to
   a) fossils b) water c) population d) soil

## Section - B

*Note: Answer any four questions.* 4x2=8

9. What is endothelium?
10. Define Genetic code.
11. Name the chemicals used in gene transfer.
12. What is cybrid?
13. What is seed ball?
14. What is called stomium?

## Section - C

*Note: Answer any three questions. Question no. 19 is compulsory.* 3x3=9

15. What are the reasons for Mendel's success in his breeding experiment?
16. Write the advantages of herbicide tolerant crops.
17. Give an account on cryopreservation.
18. Distinguish habitat and niche.
19. Write the functions of endosperm.

## Section - D

*Note: Answer the following in detail.* 2x5=10

20. a) Give a detailed account on parthenocarpy. Add a note on its significance.

(OR)

b) Describe dominant epistasis with an example.

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21. a) How is *Nicotiana* exhibit self incompatibility. Explain its mechanism.

(OR)

b) Enumerate the anatomical adaptation of xerophytes.

## PART - II (BIO - ZOOLOGY)

**(Marks : 35)**

### Section - A

*Note: i) Answer all the questions.* 8x1=8

*ii) Choose the most suitable answer from the given four alternatives and write the option code and the corresponding answer.*

1. In which mode of reproduction variations are seen
   a) Asexual b) Parthenogenesis c) Sexual d) Both (a) and (b)
2. ABO blood group in man is controlled by
   a) Multiple alleles b) Lethal genes
   c) Sex linked genes d) Y - linked genes
3. A mRNA molecule is produced by
   a) Replication b) Transcription c) Duplication d) Translation
4. Choose the correctly matched pair.
   a) Amphetamines - Stimulant
   b) LSD - Narcotic
   c) Heroin - Psychotropic
   d) Benzodiazepine - Pain killer
5. Colostrum is rich in
   a) IgE b) IgA c) IgD d) IgM
6. Pataus' syndrome is also referred to as
   a) 13 - Trisomy b) 18 - Trisomy c) 21 - Trisomy d) None of these
7. A contraceptive pill prevents ovulation by
   a) blocking fallopian tube
   b) inhibiting release of FSH and LH
   c) stimulating release of FSH and LH
   d) causing immediate degeneration of released ovum
8. ZW - ZZ system of sex determination occurs in
   a) Fishes b) Reptiles c) Birds d) All of these

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## Section - B

*Note: Answer any four questions.* 4x2=8

9. Differentiate between external fertilization and internal fertilization.
10. Draw a labeled sketch of a spermatozoan.
11. What is Lyonisation?
12. Define Antibiotic.
13. Give reasons : Genetic code is universal.
14. Who disproved Lamarck's Theory of acquired characters? How?

## Section - C

*Note: Answer any three questions. Question no. 19 is compulsory.* 3x3=9

15. What is colostrum? Write its significance.
16. How are STDs transmitted?
17. Why the human genome project is called a mega project?
18. Mention the main objections to Darwinism.
19. What are the applications of Karyotyping?

## Section - D

*Note: Answer the following questions in detail.* 2x5=10

20. a) Explain the different kinds of syngamy in living organisms.

**(OR)**

b) Explain the inheritance of sex linked characters in human being.

21. a) Write the salient features of genetic code.

**(OR)**

b) Explain the structure of immunoglobulin with suitable diagram.

♦♦♦♦♦

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